

Responses to the Comments of the Editor and Reviewers

MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

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Bowen Xie, Sheng Zhou, and Zhisheng Niu

Beijing National Research Center for Information Science and Technology,

Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

E-mail: xbw22@mails.tsinghua.edu.cn, {sheng.zhou, niuzhs}@tsinghua.edu.cn

Dear Editor and Reviewers,

We sincerely thank the Editor and the Reviewers for their careful evaluation of our manuscript and for their constructive comments. We summarize the major revisions below, followed by our detailed point-by-point responses. Changes in the marked-up manuscript are highlighted in blue.

Summary of Major Revisions

1. **Modeling and technical consistency.** Clarified the latency proxy and incorporated HO interruption with corresponding algorithm updates. Distinguished the online interference approximations from the directional interference calculations used in simulation and quantified the resulting bias in receive-rate estimates.
2. **Expanded evaluation.** Added two literature-based baselines for joint HO and BF. Reported vehicular speed distributions and analyzed how NN prediction accuracy and latency performance vary with vehicular speed. Examined the effects of gain-prediction errors on system transmit power and latency performance.
3. **Implementation overhead and presentation.** Quantified NN complexity, inference latency, and control-plane reporting overhead. Explained parameter choices, clarified the extension beyond the conference paper, expanded the discussion of related work, and added a notation table.

1 Editor

Comment 1 (Major revision decision and overall concerns)

While the work is regarded positively in general, the reviews have identified several areas that need major improvement, e.g., regarding problem formulation, technical rigor/consistency, evaluation, and presentation. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

It should be emphasized that a major revision decision does not imply eventual acceptance. A future version of the paper will only be accepted if the revision addresses the major concerns, and the reply is highly persuasive to the Editors and Reviewers.

Response from the Authors:

We thank the Editor for the opportunity to revise the manuscript and for highlighting the main concerns. We have revised the manuscript in the following three areas.

1. **Modeling and technical consistency.** We clarify the interpretation of the queueing-delay proxy and its violation probability, incorporate HO interruption into queue evolution, and update the PHO and RA algorithms accordingly. We also distinguish the interference approximations used for online decisions from the directional interference calculation with explicit RB assignments used in simulation. The combined effect of these approximations on estimated receive rates is quantified with associations, beam pairs, and RB counts held fixed.
2. **Expanded evaluation.** To broaden the comparison, we introduce two literature-based joint HO and BF baselines, O-MAPPO-adapted and MTS-GS-HBF-adapted. The former combines reinforcement learning with target-BS optimization, whereas the latter applies Gale–Shapley matching to BS–vehicle association and updates beamforming on a shorter timescale than association. We also report vehicular speed distributions and analyze how NN prediction accuracy and latency performance vary with vehicular speed. An additional sensitivity study examines how gain-prediction errors affect HO, beam probing, system transmit power, and latency performance.
3. **Implementation overhead and presentation.** To quantify implementation overhead, we report NN parameter counts, parameter storage, FLOPs, measured inference latency, and control-plane reporting overhead. We explain the algorithm parameter settings and clarify the contributions beyond the conference version and related joint optimization studies. We also add a notation table and revise terminology for consistency.

2 Reviewer 1

Comment 1 (Motivation and cross-tier heterogeneity)

The paper explains the coupling among handover, beamforming, and resource allocation in Het-Nets. It would be beneficial to further strengthen the motivation by more clearly articulating why this problem becomes fundamentally more unique and challenging in heterogeneous networks, beyond differences in system architecture. In particular, additional discussion on the new optimization challenges introduced by cross-tier heterogeneity could further improve the clarity of the problem formulation.

Response from the Authors:

We thank the Reviewer for this helpful suggestion. Cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions affect the link gains and training overhead associated with PHO and BF decisions. Together with differences in bandwidth and transmit power, they change the RB requirements and power costs of serving the same traffic, while RB budgets constrain service feasibility. Thus, association cannot be decided from link gain alone. HO also changes the loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, affecting the resource demands of other links. The revised Introduction explains how these effects motivate joint coordination of PHO, BF, and RA to reduce system transmit power under latency constraints.

Changes in the Revised Manuscript:

The revised discussion reads:

PHO and BF jointly determine the serving link gain and beam-training overhead. In HetNets, these quantities also depend on cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions. Together with differences in bandwidth and transmit power across tiers, these factors affect the number of resource blocks (RBs) required to serve a given traffic demand and the resulting power consumption, while the available RB budgets constrain service feasibility. Moreover, HO changes the traffic loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, thereby affecting the resource demands of other links. This motivates energy-efficient Coordination of handOver, Beamforming and Resource Allocation (COBRA) in vehicular HetNets.

Location in the Revised Manuscript: Pages 1–2, Section I (Introduction), paragraph beginning *PHO and BF jointly determine*.

Comment 2 (Queueing delay and end-to-end latency)

The latency formulation is currently based on queue-length constraints. The paper could be further strengthened by discussing the relationship between queueing delay and actual end-to-end latency in a two-tier vehicular HetNet. Incorporating or discussing additional latency components, such as handover interruption, beam training overhead, retransmissions, and signaling delay, would improve the realism of the latency evaluation.

Response from the Authors:

We thank the Reviewer for this suggestion. End-to-end latency includes queueing, transmission, propagation, and processing delays along the communication path. Our model focuses on queueing delay at the serving BS to assess how HO, BF, and RA jointly affect traffic accumulation through the available

service rates. As clarified in Section II-C, U measures threshold violations of the queueing-delay proxy $\tau_v(n) = q_v(n)/\lambda_v$, not a directly measured packet-delay tail probability.

We have now incorporated a first-order HO interruption model. A vehicle switching its serving BS receives no data service during the first L_{HO} slots of the next frame. Its traffic arrivals continue to accumulate in the queue during these slots, so the interruption affects the queue evolution underlying U . Beam-training overhead is already accounted for through the data-transmission fraction $1 - \eta_{m,v}^{[x]}(i)$ in the receive rate. Thus, both mechanisms affect the latency proxy through the available service rate. Their durations are therefore not added separately to the proxy.

The revised model distinguishes advance HO preparation from HO execution. HO preparation signaling is assumed to finish before the current frame ends without occupying data-service resources, whereas HO execution incurs the modeled data-service interruption. This work focuses on energy-efficient coordination of PHO, BF, and RA under queue-length-based latency constraints. To keep the modeling and analysis tractable, we assume successful handovers and do not separately model failure recovery, re-transmission delays, or other end-to-end latency components. Consistent with this scope, the manuscript evaluates a queueing-delay proxy rather than complete end-to-end latency. GAP-HO now corrects the estimated RB demand in its RB-capacity constraint to account for the slots lost to HO interruption. OTR-RA allocates no RBs to vehicles during the interruption.

Changes in the Revised Manuscript:

The explanation added before the queueing-delay proxy definition in Section II-C reads:

Queueing delay is an important component of end-to-end latency, alongside transmission, propagation, and processing delays. We focus on queueing delay to assess how HO, BF, and RA jointly affect traffic accumulation through the available service rates.

The HO assumptions added to Section II read:

After a PHO decision is made, the signaling exchanges required for HO preparation are assumed to finish before the current frame ends, without occupying data-service resources. A vehicle switching BS is assumed to experience a data-service interruption of duration $\tau_{HO} = L_{HO}\Delta t$ caused by HO execution at the start of the next frame, where L_{HO} is a nonnegative integer less than N_{HO} . We assume successful HO execution for all PHO decisions. The macro control-plane connection and periodic CSI probing remain available during this data-plane HO interruption.

The new indicator $\delta_v^{[x]}$ equals one when vehicle- v changes its serving BS between consecutive frames and zero otherwise. Equation (3) defines its data-service availability as

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{HO}\}.$$

Equation (12) and constraint C4 of **P1** now enforce

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i).$$

This makes the service rate zero during HO interruption in the existing receive-rate and queue-update equations, Eqs. (20) and (23). The latency proxy and U remain as defined in Section II-C.

Location in the Revised Manuscript: Page 3, Section II, HO assumptions and Eq. (3); Page 4, Section II-B, Eq. (12) and receive-rate model; Pages 4–5, Section II-C, paragraph preceding Eq. (24); Page 5, Section III, constraint C4 of **P1**; Page 8, Section IV-B, HO-aware RB-capacity constraint; Pages 8–9,

Section IV-D, resource allocation during HO interruption.

Comment 3 (Terminology in Fig. 1)

In Fig. 1, the use of the term “Optimal” beam-pair and gain prediction may be slightly strong given that the neural network provides estimated outputs. The presentation could be made more precise by adopting terminology such as “Predicted” or “Estimated” beam-pair/gain.

Response from the Authors:

We thank the Reviewer for this suggestion. The original term *optimal* referred to the prediction targets: the beam pair that maximizes the frame-averaged desired-link gain within the prescribed codebooks and its corresponding desired-link gain. To avoid ambiguity about prediction accuracy, we have revised the caption of Fig. 1 and the opening sentence of Section IV-A to describe predicting the best beam-pair index and its corresponding gain.

Changes in the Revised Manuscript:

The revised Fig. 1 caption reads:

Neural network architecture for predicting the next-frame best beam-pair index, its corresponding desired-link gain, and the interfering-link gain.

The opening sentence of Section IV-A now identifies the prediction targets as *best beam-pair indices, their corresponding desired-link gains, and interfering-link gains.*

Location in the Revised Manuscript: Page 6, Fig. 1 caption; Page 5, opening sentence of Section IV-A; Pages 6–7, Section IV-A, Output Prediction.

Comment 4 (Propagation of prediction errors)

The paper presents promising prediction accuracy results. It would be valuable to further analyze how prediction errors propagate to higher-layer decisions, particularly proactive handover performance. Additional experiments quantifying the impact of prediction errors on HO reliability, association accuracy, or latency performance would provide deeper insight into the robustness of the proposed framework.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added an explanation of prediction error propagation in Section IV-E and a sensitivity study in Section V-E. The study quantifies the influence of prediction errors on HO, beam probing, system transmit power and latency performance.

Gain-prediction errors change the RB demand estimates in Eq. (40), and hence the estimated power costs and RB requirements used by GAP-HO to select BS–vehicle associations. PET-BF depends on the predicted beam-pair list and search order, while its early-termination threshold depends on the predicted desired-link gain. OTR-RA uses measured desired-link gains, but errors in the predicted interfering-link gains and estimated BS RB usages affect its RB-efficiency estimates and allocation decisions. These effects propagate to the queue evolution and violation probability U through the resulting receive rates.

To quantify the effect of gain-prediction errors, we add independent zero-mean Gaussian noise in dB, with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB, to either the desired-link or interfering-link gain predictions, changing only one gain type at a time. The trained models, beam-pair candidates, and algorithm parameters remain fixed. For each traffic load and random seed, the actual channels and traffic arrivals are identical across noise settings. We rerun the MEET-COBRA framework at 9, 19, 29, and 35 Mbps

using three paired seeds, with each run covering 30 s. The case $\sigma = 0$ retains the original NN prediction errors.

At 29 Mbps, adding desired-link gain noise with $\sigma = 5$ dB increases the mean number of HOs per frame from 3.70 to 20.01 and the mean number of probes per beam search from 1.06 to 1.95. The latter statistic is averaged over the beam searches actually performed, excluding macro-BS associations and HO-interrupted slots. Additional HOs and beam probes reduce the time available for data transmission. At this operating point, U increases from 0.19% to 0.29%. These results show the influence of gain-prediction errors on MEET-COBRA, while MEET-COBRA still maintains a low violation probability at this load.

Figure 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. At 35 Mbps, however, adding noise with $\sigma = 5$ dB raises U from 1.15% to 3.84% for desired-link gain errors and 3.13% for interfering-link gain errors. Figure 11 reports the corresponding power results. Larger errors generally increase power at lower loads, whereas power remains close to the full-load level at 35 Mbps as U increases. These results show that MEET-COBRA maintains low violation probabilities under moderate additional prediction errors at light and medium loads, although HO and beam-search overhead can increase.

Changes in the Revised Manuscript:

Section IV-E adds a paragraph tracing prediction errors through GAP-HO, PET-BF, and OTR-RA to the receive rates, queue evolution, and U . The new Section V-E, *Impact of Gain-Prediction Errors*, describes the noise-injection experiment and presents U and average system transmit power in Figs. 10 and 11, respectively. It also reports the changes in HO counts and probes per beam search and explains their effect on the time available for data transmission. The Conclusion summarizes the low violation probabilities maintained at light and medium loads under moderate additional gain-prediction errors.

Location in the Revised Manuscript: Page 10, Section IV-E, final paragraph; Page 15, Section V-E and Figs. 10–11; Pages 15–16, Section VI (Conclusion).

Comment 5 (Notation table)

The notation is relatively dense, and multiple related symbols are introduced throughout the paper. Adding a notation table summarizing the main variables and symbols would significantly improve readability and make the paper easier to follow.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added Table I (*Main Notation*) to summarize the main symbols used in the system model and algorithm design. The table note explains the omission of time indices and defines tildes as indicating predicted or estimated quantities; tilde-marked variants are not listed separately when their unmarked counterparts are included.

Location in the Revised Manuscript: Page 2, Table I (Main Notation), referenced at the end of Section I.

Comment 6 (Selection and sensitivity of heuristic parameters)

Several heuristic parameters, such as ζ , M_P , and N_{iter} , are introduced in the framework. Providing additional discussion on their selection criteria, practical interpretation, or parameter sensitivity would further improve the reproducibility and robustness evaluation of the proposed method.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added the values of N_{iter} , ζ , and M_P to Table II. Section V-A briefly explains the parameter choices and their practical roles, and identifies Table II as the source of the default simulation settings.

The parameter N_{iter} specifies the number of GAP-HO iterations in each PHO frame. The first round estimates interference assuming full RB utilization at all micro BSs, and the second uses the resulting BS loads to update the RB demand estimates and BS-vehicle associations. We evaluated the GAP-HO iteration settings. Oracle channel information was used to isolate the effect of iteration settings from NN prediction errors. All configurations retain the 10 ms HO interruption, its RB-demand correction in the capacity constraints of **P2**, and the physical BS capacity limits on estimated RB usage. We compared exactly two iterations with at most three or five iterations at traffic arrival rates of 21, 27, and 35 Mbps per vehicle, using three paired traffic seeds. The latter two configurations stopped early when the maximum absolute change in estimated BS load was at most $\varepsilon = 0.1$ RB. Only the iteration settings differed between these configurations. The table reports the three-seed means of average system transmit power $\bar{P}$ (W) and violation probability U (%).

Arrival rate (Mbps)	Fixed 2 rounds ($\bar{P}$, U)	Up to 3 rounds ($\bar{P}$, U)	Up to 5 rounds ($\bar{P}$, U)
21	(34.694, 0.03241)	(34.666, 0.03234)	(34.661, 0.03227)
27	(65.996, 0.09058)	(65.948, 0.09061)	(65.951, 0.09055)
35	(174.735, 0.81714)	(174.737, 0.81235)	(174.737, 0.81235)

Across these loads, the largest power reduction was below 0.10%, and the largest reduction in U was 0.0048 percentage points. The mean iteration counts increased from 2 to 2.02–3.00 with the three-round limit and 2.02–4.16 with the five-round limit. These small performance changes relative to the additional iterations support retaining $N_{\text{iter}} = 2$ as a practical balance between performance and computational cost.

Algorithm 1 now runs a fixed number of iterations, without the convergence test or its parameter ε .

The factor ζ controls the step size for relaxing the RB capacity bounds when the continuous relaxation of **P2** is infeasible. We use $\zeta = 1.1$ to increase these bounds by 10% per step, balancing the relaxation step size against the number of solver calls. This relaxation is used to obtain an assignment before the final procedure for restoring feasibility; it does not increase the physical RB capacities.

The number of candidate beam pairs M_P determines the tradeoff between coverage of the optimal beam pair and beam search overhead. For the selected beam predictor, the list of five candidates includes the optimal beam pair in more than 99% of the predictions on the validation set. With $\eta = 2/112$, this choice limits beam search overhead to 8.93% of each slot; early termination can reduce the actual overhead. Thus, its choice is supported by prediction coverage and the associated resource cost.

Changes in the Revised Manuscript:

Algorithm 1 now performs N_{iter} rounds without a test for convergence. Section IV-B specifies “a fixed-point iteration over N_{iter} rounds.” Table II lists the number of GAP-HO iterations $N_{\text{iter}} = 2$, the RB capacity relaxation factor $\zeta = 1.1$, and the number of candidate beam pairs $M_P = 5$, consistent with the definition of M_P in Table I. It also specifies the HO interruption duration $\tau_{\text{HO}} = 10$ ms, PET-BF gain margin $G_{\text{th}} = 5$ dB, and system simulation duration of 30 s per run. The parameter discussion at the end of Section V-A reads:

We set $N_{\text{iter}} = 2$ to balance performance and computational cost, and $\zeta = 1.1$ for 10% increments in the RB capacity bounds when the continuous relaxation of **P2** is infeasible.

We use $M_P = 5$ and $G_{th} = 5$ dB to balance coverage of the optimal beam pair against beam search overhead, which occupies at most $\eta M_P = 8.93\%$ of each slot. Unless otherwise stated, the simulations use the parameter values listed in Table II.

Location in the Revised Manuscript: Pages 7–8, Section IV-B and Algorithm 1; Page 11, Section V-A, Table II and the discussion of parameter settings.

Comment 7 (Neural-network complexity and inference overhead)

The paper emphasizes the lightweight nature of the ML inference process. Including quantitative information regarding neural network complexity, such as model size, inference latency, FLOPs, or runtime overhead, would further strengthen the practicality claims of the proposed framework.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added a quantitative assessment of NN computational complexity in Section V-B. Table III lists the parameter counts and matrix FLOPs for the three predictors. Together, they contain 2,127,912 parameters, requiring 8.51 MB of FP32 parameter storage, and perform 4.20 MFLOPs of matrix computation per vehicle for stateful inference using the current-frame CSI and recurrent states.

Section IV-A clarifies that online inference retains recurrent states across frames and processes the current-frame CSI, so its computation does not grow with the length of the vehicle trajectory. To support this operation, the models undergo 100 epochs of finite-window training followed by 100 epochs of stateful fine-tuning, using truncated backpropagation through time with a truncation length of ten frames. Section V-B describes the training and validation procedures and the checkpoint-selection criteria, and the updated Fig. 5 presents both training stages.

Using the selected second-stage checkpoints, we measured three rounds of stateful inference over 16 continuous vehicle trajectories on an AMD EPYC 7742 CPU with one thread, PyTorch 2.6.0, FP32, and batch size one. Across 14,166 timed vehicle-frame inferences after warmup, the combined latency has a median of 5.38 ms and a 95th percentile of 5.72 ms. Timing includes tensor conversion, the three forward passes, state retention, top-5 index selection, and gain denormalization, but excludes CSI acquisition, preprocessing, and transmission.

Changes in the Revised Manuscript:

Section IV-A now states:

The states are retained across frames for each vehicle, so online inference processes only the current CSI feature vector.

Section V-B describes finite-window training followed by stateful fine-tuning and specifies how the beam and gain checkpoints are selected. Figure 5 shows the training and validation curves for both stages, with stars identifying the best validation value of each metric. Table III (*NN Parameter Counts and Inference FLOPs*) and the following discussion quantify NN computation:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Location in the Revised Manuscript: Page 6, Section IV-A, stateful inference; Pages 11–12, Section V-B, two-stage training, Fig. 5, Table III, and the discussion of NN computation.

3 Reviewer 2

Overall Assessment and Strengths

The joint treatment of proactive handover, beamforming, and resource allocation is a real and underexplored gap, and using the per-link resource-block demand as the common currency that links prediction, association, and scheduling is clean and well motivated. The evaluation platform is credible, since combining SUMO traces with Sionna RT ray tracing is more realistic than the stochastic channel models most comparable papers use. The algorithmic engineering is careful, with per-module complexity analysis, a Shmoys–Tardos approximation for the assignment step with a stated two-factor capacity bound, and a reasonable ablation and oracle design. Theorem 1 gives a principled starting point for the demand estimate. Overall the work is competent and the writing is mostly clear.

Response from the Authors:

We thank the Reviewer for the careful assessment of our work and the constructive comments. In the following, we respond to each comment individually and explain the corresponding revisions to the manuscript.

Comment 1 (Interference-model consistency and validation)

The interference model is crude and inconsistent with the proposed algorithms. The interference-link gain in Eq. (10) uses a Frobenius-norm average, justified by random transmit and receive beam directions, and the resource-block overlap term in Eq. (16) assumes independent and uniformly random allocation across base stations. Neither holds in the proposed system, because every interfering micro base station runs the same deterministic PET-BF, directing beams at its own vehicles, and the same deterministic OTR-RA, which places resource blocks by efficiency rather than at random. Since the SINR in Eq. (17), and therefore the entire power and latency story, rests on this approximation, please quantify how far these surrogates deviate from directional interference, for example against a beam-aware interference computation from Sionna RT, or justify why the bias is negligible at the reported loads.

Response from the Authors:

We thank the Reviewer for raising this concern. All revised system-level simulations use explicit RB assignments and the actual transmit and receive beams to compute receive rates and update queues. The Frobenius-norm-based gain predictions and expected RB overlaps guide online decisions, but are not used to compute the realized receive rates in the simulator. We have clarified the two averaging operations and quantified their combined effect on receive-rate estimates.

The directional interfering-link gain in Eq. (11) depends on the receive beam of the desired link and the transmit beam of the interfering link. For the complete, normalized DFT codebooks, $\|H\|_F^2/(M_T M_R)$ equals the gain uniformly averaged over all beam pairs. Using this average to represent the gain under the selected beams is an approximation; it does not assume random beam selection by PET-BF. Separately, OTR-RA determines RB counts, whereas each micro BS maps those counts to disjoint RB sets uniformly at random, independently of other micro BSs. Thus, deterministic RB counts are compatible with the expected overlap $k_{m,v}k_{m',v'}/K_\mu$ in Eq. (17), which averages over RB indices. Using the resulting mean interference in the rate expression remains an approximation.

For online OTR-RA decisions, Eq. (46) uses the desired-link gain measured after PET-BF, the predicted

frame-averaged interfering-link gain, and RB usage estimates from GAP-HO; instantaneous interfering-link gains are not required.

To quantify the combined approximation error, we compare the two receive-rate calculations under identical associations, beam pairs, RB counts, and Sionna RT channels along each revised MEET-COBRA run. The diagnostic uses channel-derived beam averages from Eq. (33), rather than NN predictions, to isolate the averaging approximations from prediction errors. Both calculations include beam-search overhead and HO interruption; the approximate rates are recorded only for comparison and do not alter the queues. The comparison covers all 18 arrival rates $\{1, 3, \dots, 35\}$ Mbps, with 30 s per run.

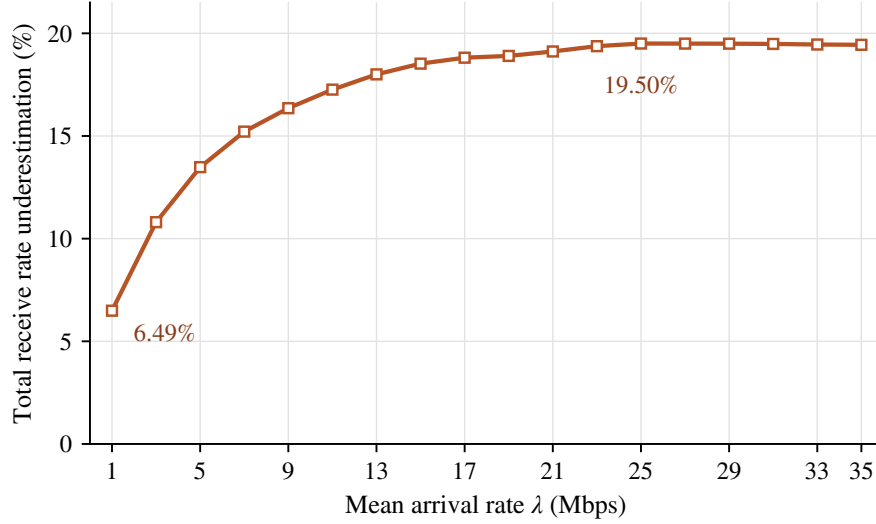


Fig. R1: Underestimation of the total receive rate of vehicles served by micro BSs when using beam-averaged gains and expected RB overlaps, relative to directional per-RB evaluation under identical decisions. Markers show means over three random seeds.

For each run, we sum receive rates over micro-BS users and average over slots. Figure R1 reports the mean underestimation across seeds at each load. The approximations underestimate this total receive rate by 6.5%–19.5% relative to directional per-RB evaluation under identical decisions, providing an empirically conservative aggregate rate estimate in the evaluated scenario.

Changes in the Revised Manuscript:

Sections II-A and II-B define the directional interfering-link gain in Eq. (11) and distinguish RB-count allocation from the random mapping to RB indices in Eqs. (17)–(18). Section IV-A defines the interfering-link gain prediction target as the frame average of the beam-averaged surrogate in Eq. (33).

Sections IV-B and IV-D specify the predicted surrogates used by GAP-HO and OTR-RA. Equation (46) and Algorithm 3 use measured desired-link gains, predicted frame-averaged interfering-link gains, and estimated RB usages. Section IV-E explains that the macro BS forwards the latter two quantities to the serving BSs before the frame begins.

Section V-A clarifies how receive rates are evaluated for all schemes:

For all schemes, receive rates are evaluated using actual RB overlaps and the selected transmit and receive beams.

Section V-D and Figs. 6–9 now report the updated system-level performance comparisons. The Abstract and Conclusion have been updated accordingly.

Location in the Revised Manuscript: Page 4, Section II-A, actual interfering-link gain and Eq. (11); Page 4, Section II-B, expected RB overlap and Eqs. (17)–(18); Page 7, Section IV-A, average gain approximation in Eq. (33); Page 7, Section IV-B, predicted frame-averaged gain surrogates; Page 9, Section IV-D, OTR-RA SINR estimate and Algorithm 3; Page 10, Section IV-E, prediction forwarding to the serving BSs; Page 11, Section V-A, receive-rate evaluation.

Comment 2 (Queue proxy versus delay guarantee)

The latency constraint is a mean-rate queue proxy, not a delay guarantee, yet the results are framed as uRLLC-grade reliability. Section II-C invokes Little’s law to write the per-slot constraint in Eq. (23). Little’s law is a long-run average identity, not a per-slot map from instantaneous backlog to instantaneous delay, so Eq. (23) bounds a scaled queue length rather than actual packet latency, and the metric U in Eq. (24) inherits this. Given the repeated uRLLC framing in the introduction, please either justify this rigorously.

Response from the Authors:

We thank the Reviewer for pointing out this distinction. We agree that the original wording did not adequately distinguish the queue-based metric from actual packet latency. Little’s law relates long-run average queue length and delay; it does not establish a per-slot equivalence between instantaneous backlog and packet delay. Section II-C now explicitly defines $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Equation (24) constrains this proxy, while U in Eq. (25) averages its threshold violations over active vehicles and time slots. Thus, U is not a directly measured packet-delay tail probability. With the HO interruption model added in response to Comment 6, interrupted slots have zero data-service rate. The resulting backlog is included in the same queueing-delay proxy and violation metric, without changing their definitions. Figure 8 and its caption now explicitly identify L_{90} and L_{99} as percentiles of the queueing-delay proxy.

We have also removed the explicit uRLLC framing from the Introduction. In the Conclusion, *latency reliability* has been replaced with *latency performance* to keep the interpretation consistent with the defined metric.

Changes in the Revised Manuscript:

The interpretation preceding Eq. (24) now reads:

Motivated by Little’s law [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

The sentence following the definition of $U_v(n)$ now reads:

Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

The corresponding sentence in the Conclusion reads:

Notably, MEET-COBRA effectively balances system transmit power and latency performance.

The explicit uRLLC framing in the Introduction has also been removed. Section V-D now states the percentile calculation, and Fig. 8 separates L_{90} and L_{99} .

Location in the Revised Manuscript: Page 1, opening paragraph of Section I; Pages 4–5, Section II-C, interpretation of Eqs. (24)–(25); Pages 13–15, Section V-D and Fig. 8, percentile calculation and results;

Pages 15–16, Section VI (Conclusion).

Comment 3 (Impact of gain-prediction error)

Gain-prediction error is not propagated into the performance metrics, and 4 dB MAE is large for mmWave link budgeting. Figure 4(b) reports validation MAE around 4.07 dB for the desired-link gain. At mmWave, 4 dB can move a link across the boundary between a usable and unusable beam, and it perturbs the demand estimate in Eq. (38) and hence U . Please add an error-propagation study, or at least plot U against injected gain-prediction noise, to show the accuracy achieved is actually sufficient.

Response from the Authors:

We thank the Reviewer for this comment. We have added an explanation of gain-prediction error propagation in Section IV-E and a sensitivity study in Section V-E. The study quantifies the effects on system transmit power and violation probability U . The revised gain predictors have validation MAEs of 3.11 and 2.51 dB for the desired-link and beam-averaged interfering-link gains, respectively.

The error-propagation discussion in Section IV-E explains the relevant decision paths. With other inputs fixed, overestimating the desired-link gain or underestimating the interfering-link gains lowers the RB demand estimate in Eq. (40), while errors in the opposite direction raise it. These estimates affect the power costs and RB requirements used by GAP-HO to select BS–vehicle associations. Desired-link gain errors also change the early-termination threshold in PET-BF. OTR-RA uses measured desired-link gains, but its RB-efficiency estimates depend on predicted interfering-link gains and estimated BS RB usages.

To quantify these effects, we add independent zero-mean Gaussian noise in dB, with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB, to either the desired-link or interfering-link gain predictions. Only one gain type is affected by this noise at a time. The trained models, beam-pair candidates, and algorithm parameters remain fixed. For each traffic load and random seed, the actual channels and traffic arrivals are identical across noise settings. We rerun MEET-COBRA at 9, 19, 29, and 35 Mbps using three paired random seeds, with each run covering 30 s. The case $\sigma = 0$ retains the original NN prediction errors.

Figure 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. At 29 Mbps, adding noise with $\sigma = 5$ dB increases U from 0.19% to 0.29% for either gain type. At 35 Mbps, the same noise level raises U from 1.15% to 3.84% and 3.13% for desired-link and interfering-link gain predictions, respectively. Figure 11 also quantifies the power cost. At 19 Mbps, adding interfering-link gain noise with $\sigma = 10$ dB increases power from 33.80 to 40.15 W while U remains approximately 0.24%. At 35 Mbps, power stays close to the full-load level while U rises. These results show that MEET-COBRA maintains low violation probabilities under moderate additional gain-prediction errors at light and medium loads.

Changes in the Revised Manuscript:

Section IV-E explains how gain errors affect RB demand estimates, PHO, beam probing, and resource allocation. Section V-E adds the separate desired-link and interfering-link gain-noise experiments. Figure 10 plots the latency constraint violation probability U , and Fig. 11 plots average system transmit power $\bar{P}$, against the standard deviation of the additional gain-prediction noise. The accompanying discussion compares the sensitivity across loads and relates it to HO interruption, beam-search overhead, and remaining RB resources. The Conclusion summarizes the low violation probabilities maintained at light and medium loads under moderate additional gain-prediction errors.

Location in the Revised Manuscript: Page 10, Section IV-E, final paragraph; Pages 11–12, Section V-B and Fig. 5(b), updated gain-prediction MAEs; Page 15, Section V-E and Figs. 10–11; Pages 15–16,

Section VI (Conclusion).

Comment 4 (Generalization of the learned models)

Generalization of the learned models is untested. The network is trained and validated on 6000 frames from a single map, a fixed four-micro-BS layout, and one traffic regime, with the split taken over non-overlapping time intervals of the same scenario, so all reported accuracy is in-distribution. There is no evaluation under a different topology, base-station placement, or arrival pattern, and the limitation is not acknowledged beyond a vague mention of online adaptation. Please add an out-of-scenario test, or state plainly that the model is site-specific and must be re-trained per deployment.

Response from the Authors:

We thank the Reviewer for highlighting this limitation. We agree that results from a single map and fixed BS layout do not establish generalization to other deployments. The revised NN training and validation sets contain disjoint vehicles from this same scenario. Following the Reviewer's suggested alternative, Section IV-A now explicitly states that deployment in a different propagation environment or BS layout requires retraining the site-specific prediction models using channel data from the target site. Generalization to other deployments has not been evaluated.

Changes in the Revised Manuscript:

Section IV-A adds the retraining requirement after the description of the site-specific model:

Deployment in a different propagation environment or BS layout requires retraining the prediction models using channel data from the target site.

Location in the Revised Manuscript: Pages 5–6, Section IV-A, opening paragraph; Page 11, Section V-B, vehicle-level training and validation split.

Comment 5 (Learning-based or optimization-based comparator)

The authors cite learning-based proactive handover and reinforcement-learning beam tracking, yet none appears as a baseline, and there is no conventional optimization comparator. A 32 percent gain over a deliberately weak baseline is not strong evidence of an advance. Please add at least one learning-based or optimization-based comparator.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added two literature-based baselines for coordinated HO and BF. One combines reinforcement learning with target BS optimization, while the other uses Gale–Shapley matching for BS–vehicle association and performs association and BF updates on different timescales.

O-MAPPO-adapted follows the reinforcement learning and optimization framework of Wang et al. [22]. A shared MAPPO actor evaluates whether each vehicle should trigger HO after every 10 m of travel. The actor and critic are initialized from models pretrained using ground-truth channel information. They are then fine-tuned using predicted channel gains across all evaluated loads to adapt the HO policy to prediction errors. Fine-tuning, validation-based checkpoint selection, and final testing use non-overlapping time intervals of the same scenario. A single checkpoint is selected based on validation performance and used at all test loads. For vehicles with active HO triggers, a mixed-integer optimizer then selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-

link and interfering-link gains as MEET-COBRA. Unlike GAP-HO, it optimizes associations only for vehicles with active HO triggers. Upon HO to a micro BS, hierarchical beam search begins after the HO interruption. It first probes the 16 combinations of eight transmit and two receive sectors, then the 16 combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, and retains the pair with the highest measured gain.

MTS-GS-HBF-adapted follows the framework of Heydarian et al. [23], which coordinates user association, analog BF, and user selection with digital BF on progressively shorter timescales. In our adaptation, Gale–Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the 32-probe hierarchical beam search and five-probe local beam tracking used by O-MAPPO-adapted. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local tracking in the remaining slots. This preserves the separation between association and BF timescales in [23].

Each BS applies OTR-RA in every slot in both baselines. Their respective beam-search overheads and the same 10 ms HO interruption are included in the evaluation.

Figs. 6–9 compare all eight schemes at arrival rates of $\{1, 3, \dots, 35\}$ Mbps per vehicle over the same 30 s mobility trace. MEET-COBRA has fewer violations than O-MAPPO-adapted at every evaluated load and lower mean power over 1–23 Mbps. O-MAPPO-adapted uses less power over 25–33 Mbps, but with substantially more violations. At 29 Mbps, MEET-COBRA uses 127.75 W with $U = 0.19\%$, whereas O-MAPPO-adapted uses 107.40 W with $U = 4.18\%$.

Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every evaluated load and fewer violations at $\lambda \geq 21$ Mbps. At 29 Mbps, MTS-GS-HBF-adapted uses 135.91 W with $U = 0.77\%$, both higher than the corresponding values for MEET-COBRA. MTS-GS-HBF-adapted has slightly fewer violations at several lower loads. Its largest advantage in mean U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA.

Section V-D also examines the HO and BF mechanisms behind these comparisons. O-MAPPO-adapted retains more traffic in the micro tier at 29 Mbps, with micro-BS RB utilization of 97.2% compared with 79.8% for MEET-COBRA. This reduces the use of higher-power macro-BS RBs but leaves less micro-BS capacity to clear queues. The discussion of MTS-GS-HBF-adapted addresses its less frequent association updates and the data-transmission time consumed by probing. These comparisons extend the evaluation beyond Reactive-OBRA to alternative learning and optimization methods for coordinated HO and BF.

Changes in the Revised Manuscript:

Section V-C introduces both baselines, explaining their core ideas, adaptations, predicted channel inputs, and common use of OTR-RA. The descriptions specify the hierarchical beam search and local beam tracking procedures. Section V-D and Figs. 6–9 compare average system transmit power $\bar{P}$, latency constraint violation probability U , queueing-delay percentiles, and macro-BS association ratios across all evaluated loads. The discussion relates these results to association decisions, RB utilization, and beam-search overhead.

Location in the Revised Manuscript: Page 12, Section V-C, benchmark descriptions; Pages 12–15,

Section V-D and Figs. 6–9, comparisons across all evaluated loads and discussion.

Comment 6 (Inference, control-plane, and handover overhead)

The paper claims lightweight inference overhead but gives no inference latency, FLOPs, or control-plane bit budget. More importantly, the central selling point of proactive handover is reduced failures and interruption, yet handover signaling and interruption latency are deferred to future work and excluded from U . Please add a like-for-like overhead comparison and at least a first-order handover-latency model.

Response from the Authors:

We thank the Reviewer for this comment. Section V-B now quantifies NN computational complexity, inference latency, and the overhead of control-plane reporting. We have also added a first-order HO interruption model and updated the PHO and RA algorithms accordingly.

1) NN computational complexity and inference latency.

Table III reports the NN parameter counts and matrix FLOPs of the beam-pair, desired-link gain, and interfering-link gain predictors. Together, the three models contain 2,127,912 parameters, requiring 8.51 MB of 32-bit floating-point (FP32) parameter storage. Their combined matrix computation is 4.20 MFLOPs per vehicle per frame. This count covers the LSTM and linear-layer matrix operations, with two FLOPs per multiply–accumulate, as specified in the table note.

Online inference processes the current-frame CSI and retains recurrent states across frames, so its per-frame computational cost does not grow with trajectory length. With FP32 inference on an AMD EPYC 7742 CPU using one thread and batch size one, the three models have a combined inference latency of 5.38 ms at the median and 5.72 ms at the 95th percentile, corresponding to 5.38% and 5.72% of the 100 ms frame duration, respectively. Timing includes tensor conversion, the three forward passes, state retention, top-5 index selection, and gain denormalization. To measure NN inference separately, the timed interval starts with preprocessed CSI and ends with the prediction outputs, excluding CSI acquisition, input preprocessing, and report transmission. The reporting overhead is quantified below.

2) Control-plane reporting budget.

For each micro-BS link, the vehicle reports the top- M_P beam-pair indices in descending order of predicted probability and two FP32 gains, one for the desired link and one for the interfering link. The full probability distribution is not uploaded. With $M = 4$, $M_P = 5$, and $M_T M_R = 256$, each index requires 8 bits, giving $4(5 \times 8 + 2 \times 32) = 416$ bits per vehicle per frame, or 4.16 kbps for 100 ms frames.

To compare reporting overhead, we consider two alternative schemes that upload CSI instead of prediction results. At the same reporting interval and 32 bits per real CSI component, uploading all complex MIMO channel matrices requires $2MM_R M_T \times 32 = 65,536$ bits per vehicle per frame (655.36 kbps). Uploading the current-frame superposed-pilot observations for centralized NN inference requires $2M_R N_P \times 32 = 4,096$ bits per vehicle per frame (40.96 kbps) with $N_P = 8$. In this second alternative, the macro BS retains recurrent states, so previous observations need not be uploaded again. The reporting overheads of the two reference schemes are 157.5 and 9.85 times that of prediction reporting, respectively.

3) HO interruption model.

The revised model distinguishes advance HO preparation from HO execution. The signaling exchanges required for advance HO preparation are assumed to be completed before the end of the current frame without consuming data-service resources. When a vehicle changes its serving BS, HO execution at the start of the next frame interrupts data service for $\tau_{HO} = L_{HO}\Delta t$ seconds, where $0 \leq L_{HO} < N_{HO}$. The

control-plane connection to the macro BS and periodic CSI probing remain available throughout this interruption.

For frame-level PHO, the revised GAP-HO accounts for the shorter service period after switching BS by applying the HO-aware correction factor $c_{m,v}^{[x+1]}$ to the estimated RB demand in the RB-capacity constraint of **P2**. This factor is equal to one if the serving BS does not change and $N_{\text{HO}}/(N_{\text{HO}} - L_{\text{HO}})$ if it changes. The correction reflects the additional RBs needed per slot during the remaining $N_{\text{HO}} - L_{\text{HO}}$ slots to serve the data accumulated during HO interruption. For slot-level RA, the revised OTR-RA excludes HO-interrupted vehicles from scheduling and assigns them zero RBs. These zero RB allocations yield zero service in the queue-update equation, while traffic continues to arrive. The resulting backlog therefore affects the queueing-delay proxy $\tau_v(n) = q_v(n)/\lambda_v$ and its violation probability U . For tractable modeling and analysis, we assume successful HO execution and do not separately model HO failure recovery, retransmission delays, or other end-to-end latency components. All system-level comparisons in Figs. 6–9 have now been rerun with this model, using $\tau_{\text{HO}} = 10$ ms for every scheme.

Changes in the Revised Manuscript:

Section IV-A clarifies that recurrent states are retained across frames, so online inference processes only the current CSI feature vector. Section V-B adds Table III (*NN Parameter Counts and Inference FLOPs*) and reports the parameter storage and measured inference latency:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Section IV-E specifies that only the ranked top- M_{P} beam-pair indices and two gain predictions per micro BS are uploaded. Section V-B quantifies this reporting overhead and compares it with the two alternative CSI-reporting schemes:

With $M = 4$, $M_{\text{P}} = 5$, and $M_{\text{T}}M_{\text{R}} = 256$, each vehicle reports, for each micro-BS link, M_{P} beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_{\text{P}} + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

Section II introduces the first-order HO interruption model, distinguishing advance HO preparation from the data-service interruption during execution. With $\delta_v^{[x]} = 1$ for a vehicle changing BS and zero otherwise, Eq. (3) defines its slot-level data-service availability as

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{\text{HO}}\}.$$

Equation (12) and constraint C4 of **P1** now enforce

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i).$$

For GAP-HO, Section IV-B revises the RB-capacity constraint of **P2**, Eq. (42), to

$$\sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m,$$

with the correction factor defined in Eq. (45) as

$$c_{m,v}^{[x+1]} = \left[1 - (1 - u_{m,v}^{[x]}) \frac{L_{\text{HO}}}{N_{\text{HO}}} \right]^{-1}.$$

The paragraph following Eq. (45) explains why vehicles switching BS require more RBs per slot after interruption. Algorithm 1 and its capacity bound are updated consistently.

For OTR-RA, Section IV-D and Algorithm 3 restrict scheduling to connected vehicles without HO interruption. Algorithm 3 initializes all $k_{m,v}^{[x]}(i)$ to zero and forms the eligible vehicle set as

$$\mathcal{V}_m^{[x]}(i) = \{v \in \mathcal{V}^{[x]} \mid u_{m,v}^{[x]} s_v^{[x]}(i) = 1\}.$$

Consequently, interrupted vehicles receive no data service in Eq. (20), while arrivals continue to accumulate through the queue update in Eq. (23). The resulting backlog affects the queueing-delay proxy and U defined in Section II-C. Table I includes the HO-related symbols. Table II specifies the HO interruption duration used in the simulations, $\tau_{\text{HO}} = 10$ ms.

Location in the Revised Manuscript: Page 2, Table I; Page 3, Section II, HO assumptions and Eq. (3); Page 4, Section II-B, Eqs. (12) and (20); Pages 4–5, Section II-C, queue evolution and latency metric; Page 5, Section III, constraint C4 of **P1**; Page 6, Section IV-A, stateful inference; Page 8, Section IV-B, Eqs. (42) and (45), and Algorithm 1; Pages 8–9, Section IV-D and Algorithm 3; Pages 9–10, Section IV-E, prediction reporting; Page 11, Section V-A, Table II, HO interruption duration; Pages 11–12, Section V-B, Table III, inference latency, and reporting budget; Pages 12–15, Section V-D and Figs. 6–9, system results with HO interruption.

Comment 7 (Traffic arrival process and heterogeneous QoS)

The traffic and QoS model is thinner than the motivation promises. The arrival process in Eq. (22) is characterized only by its mean, with no stated distribution, yet for a study whose headline metric is a violation tail, arrival burstiness dominates the result and U is hard to interpret without it. In addition, the introduction motivates heterogeneous QoS across eMBB, uRLLC, and energy, but the model carries a single backlog queue and a single latency class. Please specify the arrival process and either add a second QoS class or narrow the motivation.

Response from the Authors:

We thank the Reviewer for this comment. Section II-C now specifies Poisson arrivals with mean $\lambda_v \Delta t$, independent across vehicles and slots. This provides a reference model of independent random arrivals for comparing coordination schemes under controlled mean traffic loads. This arrival model was already used in the original simulations; the revision makes that existing assumption explicit. We also state that it does not capture temporally correlated traffic bursts. Although the algorithms do not require Poisson arrivals, performance under other arrival distributions has not been evaluated.

Section II-C now defines $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. In Eq. (25), U averages threshold violations of this proxy over active vehicles and time slots. Little's law relates long-run averages and does not imply a per-slot equivalence between this proxy and actual packet delay. Accordingly, U is not a directly measured packet-delay tail probability.

Following the Reviewer's suggestion, we have narrowed the motivation to reducing system transmit power under queue-length-based latency constraints, rather than introducing multiple QoS classes. The revised Introduction removes the explicit uRLLC framing and distinguishes the power objective from

application requirements. Although the formulation allows vehicle-specific thresholds τ_v^{th} , all simulations retain a common 20 ms threshold. Performance under heterogeneous latency requirements has not been evaluated.

Changes in the Revised Manuscript:

The arrival-process description at the end of the first paragraph of Section II-C reads:

We consider Poisson arrivals that are independent across vehicles and slots, with $a_v(n) \sim \text{Poisson}(\lambda_v \Delta t)$. This model does not capture temporally correlated traffic bursts.

The interpretation preceding Eq. (24) now reads:

Motivated by Little's law [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

The sentence following the definition of $U_v(n)$ now reads:

Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

The latency threshold is now denoted by τ_v^{th} for vehicle- v . Accordingly, Eq. (24) becomes $q_v(n) \leq \lambda_v \tau_v^{\text{th}}$, and the tiered-rounding condition in Eq. (49) becomes $q_v^{[x]}(i) > \lambda_v \tau_v^{\text{th}}/2$. The corresponding notation in the violation indicator, constraint C1 of **P1**, algorithm inputs, tables, and simulation descriptions has been updated consistently.

The traffic configuration added to Section V-A reads:

For the system-level performance comparisons in Section V-D, we vary the traffic arrival rate $\lambda_v = \lambda$ over $\{1, 3, \dots, 35\}$ Mbps and set $\tau_v^{\text{th}} = 20$ ms for all vehicles.

The application-level motivation in the Introduction now reads:

CAV applications demand high throughput for perception and infotainment and low latency for driving safety.

The subsequent phrase ‘‘Such diverse communication quality of service (QoS) requirements’’ has been changed to ‘‘These communication quality of service (QoS) requirements.’’ The system transmit-power objective is discussed separately in the following HetNet paragraph.

Location in the Revised Manuscript: Page 1, Section I (Introduction), opening two paragraphs; Page 2, Table I; Pages 4–5, Section II-C (Traffic Model), arrival-process description and interpretation of Eqs. (24)–(25); Page 5, Section III, constraint C1 of **P1**; Page 9, Section IV-D, Algorithm 3 and Eq. (49); Pages 10–11, Section V-A, traffic configuration and Table II; Pages 12–15, Section V-D, latency-result descriptions.

Comment 8 (OTR-RA acronym in Fig. 2)

Figure 2 labels the resource-allocation block OUR-RA, while the text and Algorithm 3 use OTR-RA. Please fix the acronym.

Response from the Authors:

Thank you for pointing out this typographical error. We have corrected *OUR-RA* to *OTR-RA* in Fig. 2.

Location in the Revised Manuscript: Page 9, Fig. 2, resource-allocation block.

4 Reviewer 3

Summary

This manuscript proposes MEET-COBRA, a framework for jointly coordinating handover, beamforming, and resource allocation in a two-tier vehicular network, using machine learning (LSTM) to predict channel gains and beam pairs ahead of time. The authors argue that solving this joint problem exactly is computationally impractical and instead propose a two-timescale approach: predictions drive the handover decision once per frame, while beamforming and resource allocation adapt every slot based on those predictions and on real-time measurements. The simulation setup combines SUMO mobility traces with Sionna ray-tracing channel data and the evaluation includes both oracle bounds and per-module ablations that give a fairly complete picture of where the reported gains come from. The proposed scheme shows a meaningful reduction in transmit power and latency violations compared to a reactive baseline. I have a number of concerns detailed below that should be addressed in a revision.

Response from the Authors:

We thank the Reviewer for the careful summary of our work and the constructive comments. The responses below explain how we have revised the manuscript to address the specific concerns.

Major Comments

Comment 1 (Relationship to the authors' conference paper)

This work reads as an extension of the authors' own earlier conference paper, which focused on joint handover and resource allocation for a similar vehicular HetNet setup. The framework here appears to build on that by adding beamforming as a third coordinated decision, along with a more detailed channel model and simulation setup. Given this overlap, I would encourage the authors to state clearly, early in the paper, that this is an extended version of that prior work, and to briefly summarize what is specifically new here, since it would give readers a complete picture of what this paper contributes.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added a paragraph in the Introduction that explicitly identifies this paper as an extension of our earlier conference work and summarizes the differences in channel modeling, beamforming optimization, ML prediction targets, and simulation channel data.

Changes in the Revised Manuscript:

The added paragraph reads:

This paper extends our earlier conference work on joint PHO and RA [13]. Unlike the conference version, this paper incorporates MIMO channels, beam-pair selection, and beam-training overhead into the joint optimization. We also change the ML prediction targets from vehicle positions to beam-pair indices and link gains. Moreover, the simulation dataset uses ray-traced microcell MIMO channels instead of the statistical channel model adopted in the conference version.

Location in the Revised Manuscript: Page 2, Section I (Introduction), paragraph beginning *This paper extends*.

Comment 2 (ML computation and reporting overhead versus CSI overhead)

One of the paper's central motivations is reducing the overhead of channel state acquisition, which is a reasonable goal. That said, in the proposed design, each vehicle needs to run three separate ML prediction models locally every frame, one each for beam-pair selection, desired-link gain, and interference-link gain, with all of it reported to the macro base station each frame. This introduces its own computational and reporting overhead, and it isn't clear from the paper how that overhead compares to the CSI overhead the framework is designed to avoid in the first place. A brief discussion of this would strengthen the paper's claim of achieving lower overhead with lightweight ML inference.

Response from the Authors:

We thank the Reviewer for this suggestion. We agree that the computational and reporting costs should be quantified alongside the reduction in CSI acquisition overhead. Section V-B now reports these costs and compares prediction reporting with alternative CSI-reporting schemes.

For local computation, the three predictors together contain 2,127,912 parameters, require 8.51 MB of 32-bit floating-point (FP32) parameter storage, and perform 4.20 MFLOPs of matrix computation per vehicle per frame, as reported in Table III. They retain recurrent states across frames and process only the current-frame CSI, so per-frame computational complexity does not grow with trajectory length. Using FP32 arithmetic, a batch size of one, and a single thread on an AMD EPYC 7742 CPU, the three models have a combined inference latency with a median of 5.38 ms and a 95th percentile of 5.72 ms, compared with the 100 ms frame duration.

For CSI acquisition, the pilot-superposition procedure in Section IV-A uses $N_p = 8$ probing rounds per frame, with all $M = 4$ micro BSs transmitting simultaneously in each round. Under the same receiver observation model, exhaustively sweeping the $M_T = 32$ transmit beams of each BS separately would require $MM_T = 128$ rounds. Thus, pilot superposition reduces the number of frame-level CSI probing rounds by a factor of $MM_T/N_p = 16$.

For reporting, each vehicle sends, for each micro-BS link, the top- M_p beam-pair indices in descending order of predicted probability and two FP32 gains. The full probability distribution is not uploaded. With $M = 4$, $M_p = 5$, and $M_T M_R = 256$, the reporting overhead is $M(8M_p + 2 \times 32) = 416$ bits per vehicle per frame, or 4.16 kbps. For comparison, two alternatives upload CSI instead of predictions. At the same reporting interval and 32 bits per real CSI component, full MIMO-CSI reporting requires $2MM_R M_T \times 32 = 65,536$ bits, whereas current-frame superposed-pilot observations for centralized inference require $2M_R N_p \times 32 = 4,096$ bits per vehicle per frame. In the latter case, the macro BS retains recurrent states, so past observations need not be uploaded again. The reporting overheads of the two reference schemes are 157.5 and 9.85 times that of prediction reporting, respectively.

Overall, MEET-COBRA reduces the overhead of frame-level CSI probing and uplink reporting at the cost of modest local computation per vehicle.

Changes in the Revised Manuscript:

Section IV-A describes pilot superposition and stateful inference. Table III in Section V-B gives the parameter counts and matrix FLOPs of the three predictors, followed by:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference

using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Section IV-E specifies that only the ranked candidate beam-pair indices and two gain predictions per micro BS are uploaded. Section V-B adds the reporting-budget comparison:

With $M = 4$, $M_P = 5$, and $M_TM_R = 256$, each vehicle reports, for each micro-BS link, M_P beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_P + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

Location in the Revised Manuscript: Page 6, Section IV-A, Input Layer and Shared Backbone; Pages 9–10, Section IV-E, prediction reporting; Page 11, Table II, pilot and antenna configuration; Pages 11–12, Section V-B, Table III, inference latency, and reporting budget.

Comment 3 (Coordinated handover and beamforming baseline)

The comparison in Section V is against a reactive baseline, and that baseline is fairly weak on all three fronts at once: handover triggered by signal-strength thresholds, beam selection through random probing, and resource allocation under a worst-case interference assumption. The per-module ablation study shown in the paper is a nice complement to this, since it shows what each of the three proposed modules is individually contributing. It would also help to compare against a coordinated handover and beamforming baseline from existing literature.

Response from the Authors:

We thank the Reviewer for this suggestion. We have retained the module ablations and added two literature-based baselines for coordinated HO and BF. One combines reinforcement learning with target BS optimization, while the other uses Gale–Shapley matching for BS–vehicle association and performs association and BF updates on different timescales.

O-MAPPO-adapted follows the reinforcement learning and optimization framework of Wang et al. [22]. A shared MAPPO actor evaluates whether each vehicle should trigger HO after every 10 m of travel. The actor and critic are initialized from models pretrained using ground-truth channel information. They are then fine-tuned using predicted channel gains across all evaluated loads to adapt the HO policy to prediction errors. Fine-tuning, validation-based checkpoint selection, and final testing use non-overlapping time intervals of the same scenario. A single checkpoint is selected based on validation performance and used at all test loads. For vehicles with active HO triggers, a mixed-integer optimizer then selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-link and interfering-link gains as MEET-COBRA. Unlike GAP-HO, it optimizes associations only for vehicles with active HO triggers. Upon HO to a micro BS, hierarchical beam search begins after the HO interruption. It first probes the 16 combinations of eight transmit and two receive sectors, then the 16 combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, retaining the pair with the highest measured gain.

MTS-GS-HBF-adapted follows the framework of Heydarian et al. [23], which coordinates user association, analog BF, and user selection with digital BF on progressively shorter timescales. In our

adaptation, Gale–Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the 32-probe hierarchical beam search and five-probe local beam tracking used by O-MAPPO-adapted, without using predicted beam-pair candidates or PET-BF. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local tracking in the remaining slots. This preserves the separation between association and BF timescales in [23].

Each BS applies OTR-RA in every slot in both baselines. Their respective beam-search overheads and the same 10 ms HO interruption are included in the evaluation.

Figs. 6–9 compare all eight schemes at arrival rates of $\{1, 3, \dots, 35\}$ Mbps per vehicle over the same 30 s mobility trace. Across the evaluated range, MEET-COBRA has fewer violations than O-MAPPO-adapted at every load and lower mean power over 1–23 Mbps. O-MAPPO-adapted uses less power over 25–33 Mbps, but with substantially more violations. Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every load and fewer violations at $\lambda \geq 21$ Mbps. MTS-GS-HBF-adapted has slightly fewer violations at several lower loads. Its largest advantage in mean U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA.

Section V-D also examines the HO and BF mechanisms behind these comparisons. O-MAPPO-adapted retains more traffic in the micro tier at 29 Mbps, with micro-BS RB utilization of 97.2% compared with 79.8% for MEET-COBRA. This reduces the use of higher-power macro-BS RBs but leaves less micro-BS capacity to clear queues. The discussion of MTS-GS-HBF-adapted addresses its less frequent association updates and the data-transmission time consumed by probing. The added comparisons therefore evaluate alternative HO and BF coordination methods beyond Reactive-OBRA.

Changes in the Revised Manuscript:

Section V-C introduces both baselines, explaining their core ideas, adaptations, predicted channel inputs, and common use of OTR-RA. The descriptions specify the hierarchical beam search and local beam tracking procedures. Section V-D and Figs. 6–9 compare average system transmit power $\bar{P}$, latency constraint violation probability U , queueing-delay percentiles, and macro-BS association ratios across all evaluated loads. The discussion relates these results to association decisions, RB utilization, and beam-search overhead.

Location in the Revised Manuscript: Page 12, Section V-C, O-MAPPO-adapted and MTS-GS-HBF-adapted; Pages 12–15, Section V-D and Figs. 6–9, multi-seed performance comparisons.

Comment 4 (Vehicle-speed distribution and mobility sensitivity)

Despite high-mobility vehicular scenarios being the paper’s central motivation, it does not report any information on vehicle speed anywhere in the manuscript. Speed is arguably one of the main factors driving how quickly channel conditions change from one frame to the next, so it directly affects how difficult the prediction problem actually is. The authors do state that training and validation samples are drawn from non-overlapping time intervals of the same simulated traffic, but the vehicle speed distribution within that dataset is never reported. It would help to see the speed distribution used, and a result showing how performance of the proposed framework changes across different mobility levels.

Response from the Authors:

We thank the Reviewer for this suggestion. We agree that vehicle speeds are needed to interpret both prediction accuracy and system performance. We have added the speed distributions of the training, validation, and system evaluation datasets, together with prediction and latency results for four speed groups.

1) Vehicle-speed distributions.

Figure 4 in Section V-A now reports the speed CDFs, reproduced below in Fig. R2. The statistics are calculated over vehicle-frame observations, with stationary samples defined by zero speed.

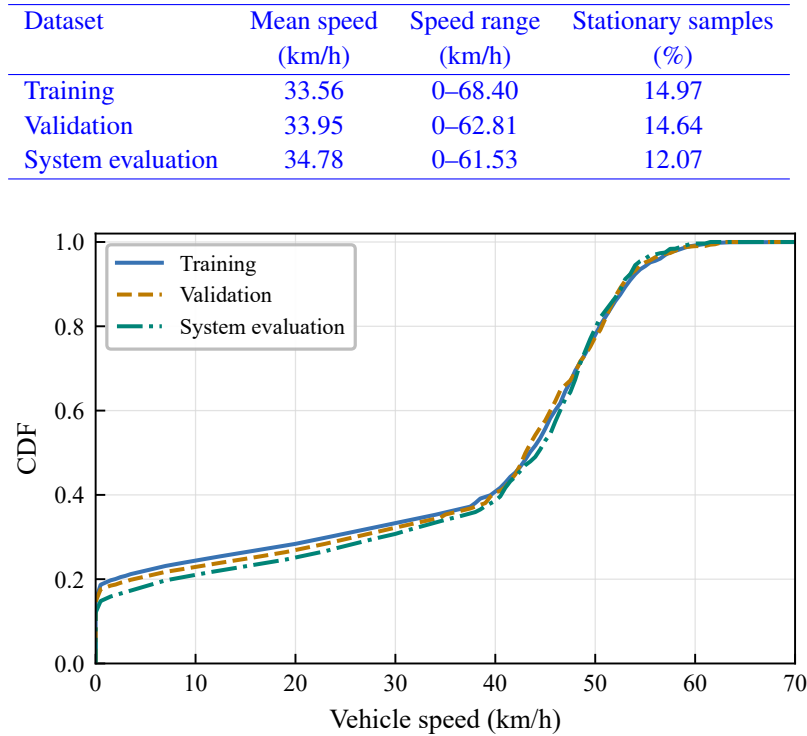


Fig. R2: Vehicle-speed CDFs over vehicle-frame samples from the training, validation, and system evaluation datasets.

2) Prediction performance across speed groups.

We use four predefined speed groups of $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h. The first group includes stationary and nearly stationary vehicles. The same selected checkpoints are used for all groups, with recurrent states carried along each complete vehicle trajectory. Predictions are grouped afterward by vehicle speed in the target frame. The following results use the prediction outputs from the system evaluation dataset, without additional training. Metrics are averaged over the four micro-BS links and all samples in each group. The gain metrics refer to the optimal desired-link gain and the beam-averaged interfering-link gain.

Speed group (km/h)	Beam accuracy (%)		Gain MAE (dB)	
	Top-1	Top-5	Desired	Interfering
$[0, 1)$	96.06	99.74	1.13	1.19
$[1, 20)$	87.92	99.42	2.31	1.79
$[20, 40)$	87.23	99.18	3.15	2.32
≥ 40	87.61	99.57	3.73	3.07

Both gain MAEs increase across the speed groups. Top-1 beam accuracy is lower in the three higher-speed groups than in the $[0, 1)$ km/h group, but top-5 accuracy remains above 99% in every group. Thus, the list of five candidate beam pairs retains high coverage of the optimal beam pair despite the larger gain errors in the higher-speed groups. Section V-B now reports the four pairs of gain MAEs and the top-5 accuracy result.

3) System latency performance across speed groups.

We also analyzed MEET-COBRA and Oracle-MC at all 18 arrival rates $\{1, 3, \dots, 35\}$ Mbps with three random seeds over the same 30 s urban mobility trace. Grouping is applied after simulation, preserving the original trajectories and queue histories. Within each run and speed group, L_{99} is the 99th percentile of the vehicle-slot samples of the queueing-delay proxy $\tau_v(n)$. The reported values are averaged over the three runs.

For MEET-COBRA, the mean L_{99} remains below 20 ms in all four groups at every evaluated arrival rate up to 33 Mbps. At 35 Mbps, the four values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. Section V-D now reports these results, showing both the latency performance across the observed speed range and the larger delays at the highest load. In the ≥ 40 km/h group at this load, Oracle-MC also has $L_{99} = 26.61$ ms, so the observed latency violations cannot be attributed entirely to prediction errors.

Changes in the Revised Manuscript:

Section V-A adds Fig. 4 and reports the mean speed and stationary sample fraction for each dataset, together with the speed range of the system evaluation dataset. Section V-B adds the four speed groups and their gain MAEs, noting that top-5 beam prediction accuracy exceeds 99% in every group.

Section V-D adds the following analysis to the discussion of latency percentiles:

When evaluated separately for the speed groups $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h, MEET-COBRA maintains a mean L_{99} below 20 ms in every group at all evaluated arrival rates up to 33 Mbps. At 35 Mbps, the corresponding values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. These results indicate that MEET-COBRA maintains low latency across the observed speed groups over most evaluated loads, while larger delays arise in the higher-speed groups at the highest load.

Location in the Revised Manuscript: Pages 10–11, Section V-A and Fig. 4, speed distributions; Page 11, Section V-B, prediction performance by speed group; Page 14, Section V-D, latency performance by speed group.

Minor Comments

Comment 5 (OTR-RA acronym in Fig. 2)

Fig. 2 labels the resource allocation block “OUR-RA,” while the surrounding text and Algorithm 3 consistently use “OTR-RA.”

Response from the Authors:

Thank you for pointing out this typographical error. We have corrected *OUR-RA* to *OTR-RA* in Fig. 2.

Location in the Revised Manuscript: Page 9, Fig. 2, resource-allocation block.

Comment 6 (Handover-algorithm parameters in the simulation table)

A few parameters used in the handover algorithm aren't listed in the main parameter table: N_{iter} (the maximum number of iterations in the fixed-point refinement step), ζ (the factor used to relax the capacity constraint when the relaxation is infeasible), and ε (the convergence tolerance for that iteration) in Algorithm 1. Worth adding for reproducibility.

Response from the Authors:

We thank the Reviewer for pointing out the missing parameter settings. Table II now lists $N_{\text{iter}} = 2$ and $\zeta = 1.1$, and Section V-A explains their use. Algorithm 1 now performs exactly N_{iter} rounds without a convergence test, so ε has been removed.

Changes in the Revised Manuscript:

Table II adds the number of GAP-HO iterations $N_{\text{iter}} = 2$ and the RB capacity relaxation factor $\zeta = 1.1$. Section IV-B and Algorithm 1 specify N_{iter} rounds and remove the convergence test and its parameter ε . Section V-A explains the parameter choices and states that Table II gives the default simulation settings.

Location in the Revised Manuscript: Pages 7–8, Section IV-B and Algorithm 1; Page 11, Section V-A, Table II and the discussion of parameter settings.

Comment 7 (Related work on joint optimization)

The related work section covers handover, beamforming, and resource allocation as separate lines of work, even though the paper's central claim rests on coordinating them. It would strengthen the positioning of the contribution to also discuss prior work that combines two or more of these problems.

Response from the Authors:

We thank the Reviewer for this helpful suggestion. We have added a paragraph to the Introduction reviewing three studies that coordinate HO with beam tracking, HO with beam selection and bandwidth allocation, and user association with hybrid BF and queue-aware scheduling, respectively. The discussion acknowledges these existing coordination approaches and distinguishes our objective of reducing system transmit power through joint PHO, BF, and RA while meeting latency requirements. To support this positioning, we have also clarified the energy-saving motivation in the preceding HetNet discussion by explaining the balance between transmit power and timely traffic service.

Changes in the Revised Manuscript:

The added discussion of joint optimization reads:

Several studies also coordinate two or more of these decisions. Khosravi et al. [21] use deep Q-learning to coordinate HO and beam tracking, accounting for beam-training overhead in the effective throughput. Wang et al. [22] combine multi-agent reinforcement learning for HO triggering with optimization of target BS and beam selection and bandwidth allocation to improve throughput and delay performance. Heydarian et al. [23] coordinate user association, hybrid BF, and queue-aware user scheduling on multiple timescales. However, these studies primarily target throughput and delay performance, and do not explicitly address the joint coordination of PHO, BF, and RA for reducing the system transmit power while meeting latency requirements.

The energy-saving motivation earlier in the Introduction now reads:

Such HetNet deployments can also reduce system transmit power through traffic-aware resource allocation. Realizing these benefits requires balancing transmit power against timely traffic service. Increasing resource usage can reduce delay but also raise power consumption [6].

The three newly added references are listed below, using their numbers in the revised manuscript:

- [21] S. Khosravi, H. S. Ghadikolaie, J. Zander, and M. Petrova, "Reinforcement Learning-based Joint Handover and Beam Tracking in Millimeter-wave Networks," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2023, pp. 1–6.
- [22] R. Wang, Y. Sun, C. Zhang, B. Yang, M. Imran, and L. Zhang, "A novel handover scheme for millimeter wave network: An approach of integrating reinforcement learning and optimization," *Digital Commun. Netw.*, vol. 10, no. 5, pp. 1493–1502, 2024.
- [23] M. Heydarian, D. Colle, M. Pickavet, and W. Tavernier, "Multi-timescale scheme for cooperative user association and hybrid beamforming in mmWave MIMO systems," *EURASIP J. Wireless Commun. Netw.*, vol. 2025, Art. no. 55, 2025.

Location in the Revised Manuscript: Page 1, Section I (Introduction), paragraph beginning *Several studies also coordinate*; Page 1, HetNet paragraph beginning *In practice*; references [21]–[23].

Yours sincerely,
Bowen Xie, Sheng Zhou, and Zhisheng Niu